MIL-M-13849 (Ord)
14 December 1954
SUPERSEDING
WXS-198
2 November 1948

MILITARY SPECIFICATION

MOTOR, TRANSMISSION DRIVE, FOR FUZE
SETTER RAMMER, COMBINATION M20

1. SCOPE

1.1 Scope.- This specification covers a special motor with a high starting torque capacity for use with Setter-Rammer, Combination, Fuze, M20.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MIL-G-10924	-Grease, Automotive and Artillery
MIL-P-16298	-Preservation, Packaging, Packing, and Marking of Electric Machines having Rotating Parts (Includes Associated Repair Parts)

U. S. ARMY

51-2	-Artillery Carriages, Mounts and Vehicles, General Specifications Governing the Manufacture and Inspection of

STANDARDS

MILITARY

MIL-STD-108	-Definitions of and Basic Requirements for Enclosures for Electric & Electronic Equipment.

DRAWING

ORDNANCE CORPS

D-49836	-Motor

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards
Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

National Electrical Manufacturers Association
Electrical Manufacturers Association Electrical Standards

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Material.- All parts of motor shall be of rugged construction and of materials best suited for the purpose intended and to withstand extreme heat, cold, and shock of gun fire.

3.2 Electrical.

3.2.1 Rating.- Motor shall be rated at 3 H.P. 110 Volts, 3-phase, 60 cycles.

3.2.2 Speed.- Motor shall have a synchronous speed of 1800 R.P.M. Speed shall not reduce below 1600 R.P.M. during a peak load of 4 H.P.

3.2.3 Motor characteristics.- Motor current, speed and torque characteristics shall be similar to figure 1.

3.2.4 Motor acceleration.- Motor shall accelerate the rammer fly-wheel before the starting current damages the windings since this motor does not have thermal overload protection.

3.2.5 Motor torque.- Motor starting torque shall be such that the starting current demand for accelerating the flywheel will drop to idle load requirements immediately.

3.2.6 Temperature.- Motor shall operate at ambient temperatures from -40°F to 120° F.

3.3 General.

3.3.1 Bearings.- Motor shall be equipped with ball bearings.

3.3.2 Grease fittings.- Grease openings shall be plugged with pipe plugs. American National taper pipe threads shall be conforming to Handbook H28.

3.3.3 Lubrication.- Ball bearings shall be pre-lubricated with grease conforming to Specification MIL-G-10924.

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3.3.4 Motor enclosure.- Motor and terminal box shall be watertight as specified in Standard MIL-STD-108, and shall be provided with a drain and plug in the lowest part of the enclosure for drainage of moisture.

3.3.5 Fittings.- All motor fittings shall be galvanized so as to render them adequately resistant to corrosion.

3.3.6 Motor drive.- A double "V" belt drive shall be used between the motor and the rammer flywheel shafts.

3.4 Operation and duty cycle.- The transmission drive shall be capable of operating satisfactorily under the following conditions:

3.4.1 Idle load.- The motor shall operate the ammunition feed shaft at low speed with the motor load at approximately 1.2 horsepower. The motor shall be capable of operating under idle load conditions for periods of several hours without damage.

3.4.2 Full load.- The motor shall operate the ammunition feed shaft at high speed with a motor load of approximately 1.8 horsepower.

3.4.3 Ramming rolls.- When ramming rolls are running at high speed, a minimum peak load of 4.0 horsepower shall be required to ram the shell into the gun.

3.4.4 Ramming operation.- After the ramming operation the motor load shall be reduced to 1.2 horsepower.

3.4.5 Ramming time.- The total time for one cycle shall be 1/2 second.

3.4.6 Emergency operating conditions.- The rammer shall be capable of operating 100 consecutive times at the rate of 25 times in 1-minute.

3.4.7 Normal operating conditions.- The rammer shall be capable of operating once every 3 seconds for 10 to 15 consecutive operations.

3.4.8 Motor.- Motor shall perform in any position and shall meet all requirements of this specification.

3.5 Workmanship.- Standards of workmanship shall be such that equipment will meet the requirements of 3.1 through 3.4.8. All components, subassemblies, parts, and the final assembly shall meet the inspection requirements for defects in quality and dimensions as specified in 4.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection.- Each motor shall be inspected for conformance with this specification and Specification 51-2.

4.2 Fuze setter rammer performance test.- Motor shall be installed with Setter Rammer, Combination Fuze, M20, and tested to insure that the requirements of this specification are fulfilled.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing, and marking shall be in accordance with Specification MIL-P-16298.

6. NOTES

6.1 Ordering data.- Contracts and invitations for bids should specify the title, number, date of this specification, and type of packing.

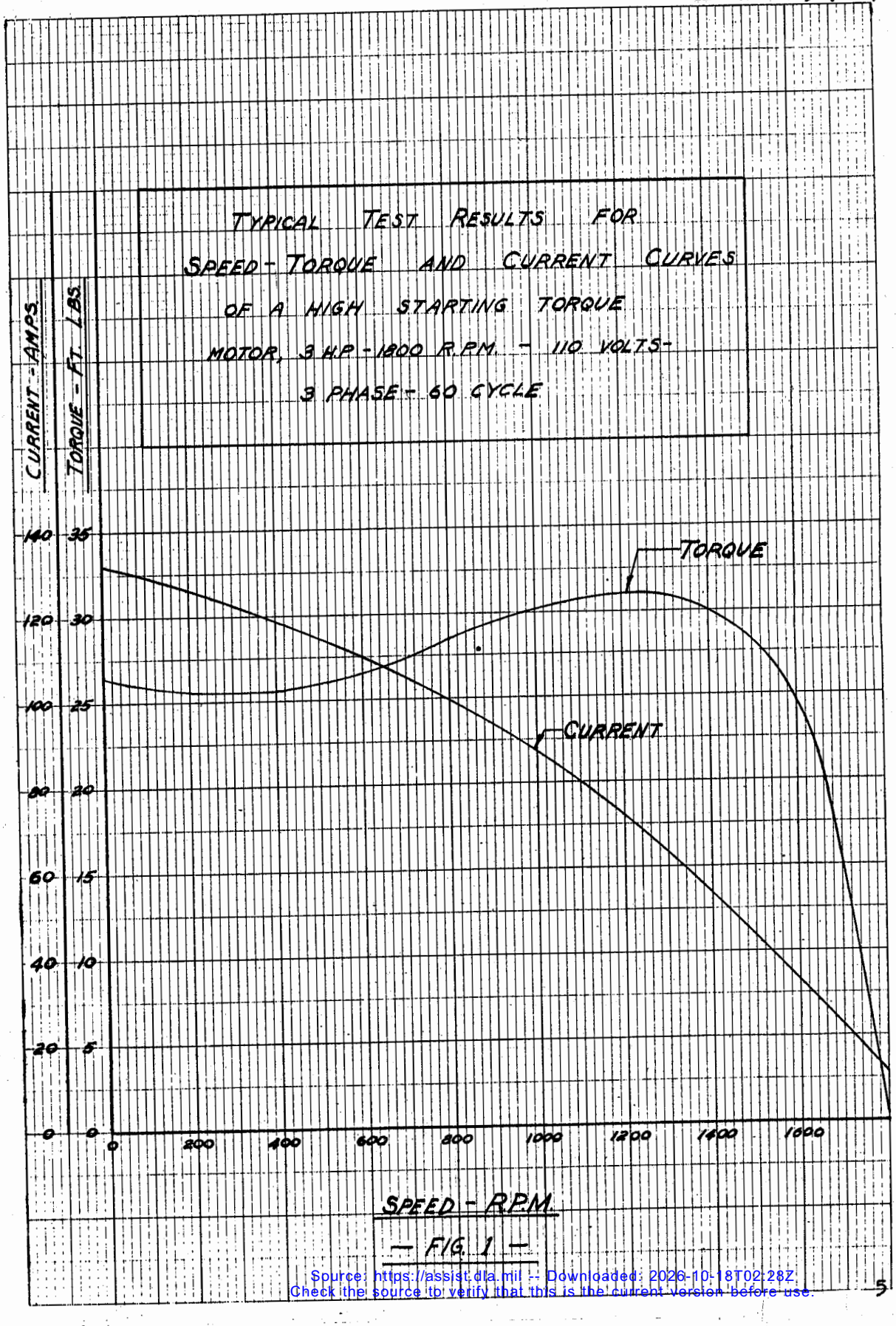
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— FIG. 1 —

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